

Pneumonia Diagnosis Detection and Localization

Project 34 | Chest X-ray classification with DenseNet121 and Grad-CAM localization | [Your name, roll number] | October 2026

1. Problem Statement

Pneumonia is commonly diagnosed from chest X-rays, but reading them takes expert time. This project builds a deep-learning system with two outputs: (i) **detection**, a pneumonia / normal prediction with a confidence score, and (ii) **localization**, a heatmap and region showing where in the X-ray the model found evidence of pneumonia. The localization is learned without any location labels during training, as in the Stanford CS229 reference project. The system is an educational prototype and not a clinical diagnostic tool.

2. Dataset

We used the RSNA Pneumonia Detection Challenge dataset (Kaggle): chest X-rays in DICOM format with a binary label (Target = 1 pneumonia, 0 not) and, for positive cases, ground-truth bounding boxes. Some patients have more than one box. We kept one row per X-ray and made a stratified **70% / 15% / 15%** train / validation / test split, which preserves the class ratio in each part. The pneumonia class is the minority (about one in four to five images). Evaluation used a fixed random sample of **500 test images (106 pneumonia, 394 normal)**. The bounding boxes were used only to evaluate localization, never for training.

3. Approach

Preprocessing. Each DICOM image is rescaled to 0-255, converted to 3-channel RGB, resized to 224x224 and normalized with ImageNet statistics. **Model.** An ImageNet-pretrained DenseNet121 (transfer learning) with its final layer replaced by a single output unit (pneumonia logit). **Class imbalance.** Weighted binary cross-entropy (*pos_weight* = negatives / positives) so the model cannot score well by always predicting normal. **Localization.** Grad-CAM on the last DenseNet feature map (7x7): each feature map is weighted by its average gradient with respect to the pneumonia output, summed, passed through ReLU, upsampled and scaled to [0,1]. Thresholding the heatmap gives a binary region, which is compared with the ground-truth boxes by **Intersection over Union (IoU)**.

4. Implementation

Everything was implemented in PyTorch in a Google Colab notebook (T4 GPU). Two training runs were made:

	Baseline	Final model
Training data	3,000 images (subset)	Full training split (about 18k images)
Augmentation	None	Horizontal flip, 10° rotation, brightness/contrast jitter
Optimizer	Adam, lr 1e-4	Adam, lr 3e-5, weight decay 1e-4, ReduceLROnPlateau
Epochs / stopping	5 epochs; best checkpoint = epoch 1	Early stopping (patience 3); stopped at epoch 6; best = epoch 3
Best validation AUC	0.8526	0.8806

The baseline **overfit**: training loss fell from 0.83 to 0.22 while validation loss rose from 0.78 to 1.23 and validation AUC dropped after epoch 1. The checkpoint with the best validation AUC was kept in both runs. For the final model, more data, augmentation, a lower learning rate and weight decay kept validation loss roughly flat (about 0.67 to 0.70) and raised validation AUC.

5. Results

All numbers are on the same 500-image test set, evaluated once, with a classification threshold of 0.5.

Metric	Baseline	Final model
ROC-AUC	0.852	0.884
Accuracy	0.772	0.808
Precision	0.477	0.531
Recall (sensitivity)	0.792	0.802
F1-score	0.596	0.639
Mean IoU (Grad-CAM threshold 0.5)	0.218	0.279

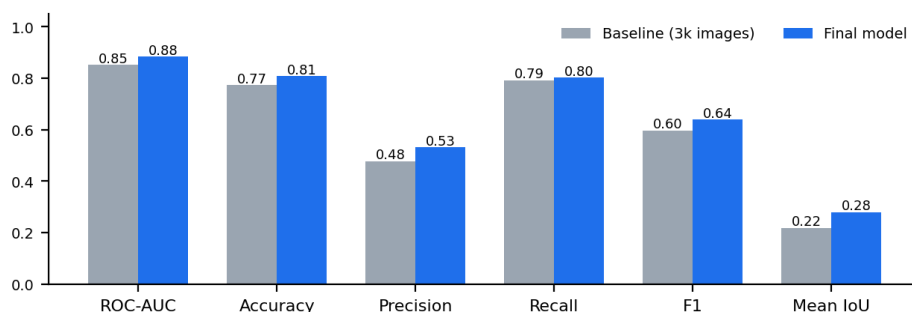


Figure 1. Baseline vs final model on the test set.

Detection. The final model's confusion matrix is TN = 319, FP = 75, FN = 21, TP = 85 (specificity 0.810). It finds 85 of the 106 pneumonia cases, but also raises 75 false alarms on 394 normal images, which is why precision is only 0.53. In a screening setting a false negative is usually the costlier error, so the threshold could be lowered to raise recall at the cost of precision.

Localization. Over the 106 pneumonia cases, mean IoU was 0.261, 0.274, 0.279, 0.267 and 0.247 at heatmap thresholds 0.3, 0.4, 0.5, 0.6 and 0.7 (baseline: 0.213, 0.222, 0.218, 0.203, 0.176). In the *pointing game*, the single hottest heatmap pixel fell inside a ground-truth box in **47.2%** of cases. Heatmaps are visibly blocky because they come from a 7x7 grid. Qualitatively, on test images chosen for the demo, the heatmap often covered the annotated lung region (best-case IoU about 0.6), but missed cases show attention on the heart and diaphragm rather than the lung fields, which suggests the model partly uses non-lung cues. One missed case had $P(\text{pneumonia}) = 0.385$, just under the cutoff. The demo examples were selected as the best-localized detections, so they are not representative of average performance.

6. Conclusion

A transfer-learned DenseNet121 detects pneumonia on RSNA chest X-rays with test ROC-AUC 0.884 and recall 0.80, and Grad-CAM gives approximate localization (mean IoU 0.28, pointing-game hit rate 47%) without any location supervision. Moving from a 3k-image baseline to the full training set with augmentation and regularization improved every metric and reduced overfitting. **Limitations:** a single train/validation/test split with no cross-validation; evaluation on a 500-image test sample, so estimates are uncertain; moderate precision; Grad-CAM resolution of 7x7; no external validation on other hospitals' data; not a clinical tool. **Future work:** evaluate on the full held-out test split, use higher-resolution CAM methods or a true detector (e.g., Faster R-CNN) with lung segmentation as in the Stanford project, tune the decision threshold for sensitivity, and validate externally.